

Lab 1 – Spoiler Alert Product Description

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CS 411W, Spring 2026

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February 13, 2026

Lab 1 Draft – Version 2

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1. Introduction

Food recalls are an ongoing public health concern, affecting millions of consumers every year. According to the U.S. Food and Drug Administration (FDA) and the U.S. Department of Agriculture (USDA), hundreds of recall incidents occur annually due to contamination, undeclared allergens, or foreign materials in products (University of Nebraska-Lincoln, 2025). Despite the frequency and seriousness of these events, many consumers remain unaware of active recalls until it is too late. This lack of awareness contributes to preventable illnesses and even deaths caused by unsafe food consumption (World Health Organization, 2024).

The core issue lies not in the absence of recall information but in how it is distributed. Current recall communication systems rely heavily on official government websites and static email alerts, which are often overlooked by the public. Studies show that only 13 percent of Americans regularly check FDA or CDC sites for recall updates, and fewer than 3 percent subscribe to automatic alerts (Gravalese, 2025). This information gap leaves the most vulnerable populations — such as children, the elderly, and those with allergies or weakened immune systems — at increased risk of exposure to contaminated products.

Three main gaps exacerbate the problem. First, an awareness gap exists because recall notices spread slowly and unevenly, leaving many consumers uninformed. Second, an engagement gap persists as consumers often underestimate the urgency of recalls. Finally, a customization gap remains because existing systems lack personalized alerts tailored to individual user needs.

A modern solution must be centralized, personalized, immediate, and trustworthy. It should gather verified recall data directly from government sources and deliver it through user-preferred channels (such as push notifications or email) in clear, concise language. Users should be able to filter recalls by allergens, track their inventory, and receive real-time alerts that help them act quickly. By combining accessibility with customization, such a platform can motivate engagement and enhance safety.

Spoiler Alert is a mobile application designed to bridge these gaps. The application provides personalized, immediate, and dependable food recall alerts to help consumers and families avoid unsafe products. Features include barcode and receipt scanning, custom allergen

filters, and real-time push notifications from FDA and USDA data. By bringing clarity, speed, and trust into recall communication, Spoiler Alert aims to ensure that no consumer is left unaware — ensuring that consumers are promptly informed of any recalls relevant to their purchased products.

2. Product Description

2.1 Key Product Features and Capabilities

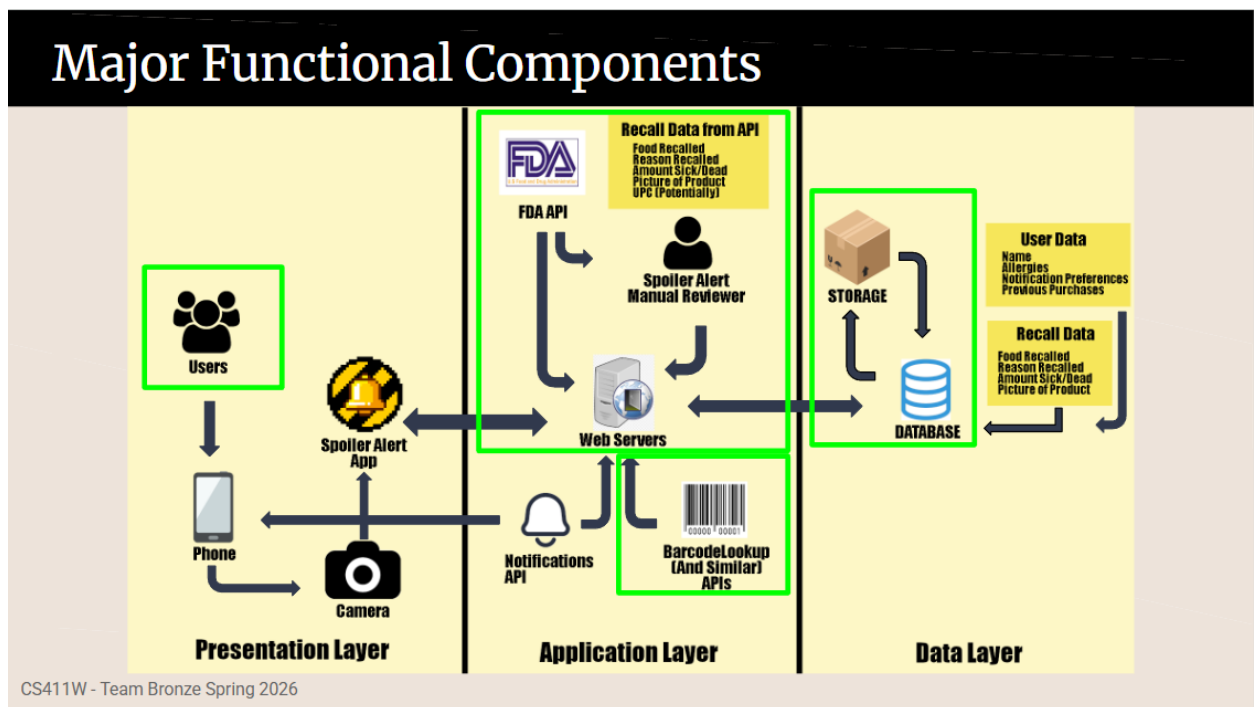
- Spoiler Alert streamlines recall awareness with multiple methods of notification.
- Spoiler Alert has functionality to specialize the type of reports based on factors like:
Allergies and Location
- The user can scan receipts and barcodes for Spoiler Alert to track what recalls the user may be at risk of.
- Provide real-time recall notifications via email and mobile app alerts.
- Aggregates recall data from FDA, USDA, and CPSC databases to ensure complete and accurate information.
- Offers a personalized inventory summarizing all active recalls affecting the user's stored products or scanned items.
- Allows users to filter recalls by type or category (food, cosmetics, and medicine).
- Utilizes cloud-based data synchronization so that updates reach users across all their devices.
- Integrates a search and filter function for recalls by product name, brand, or UPC.

2.2 Major Components (Hardware/Software)

- App as the main user interface.
- Spoiler Alert Servers:

- Our servers grab data from the FDA API.
- Our Servers try to pull information from the API to alert users about the recall.
- Our servers will save previous user purchases to the servers to alert them about future recalls.
- In the case that a UPC cannot be gathered from the recall, a Spoiler Alert employee will manually gather all relevant UPCs and link them.
- Include MFCD from the slides?

Figure 1: Major Functional Components



3. Identification of Case Study

The primary focus for the Spoiler Alert case study represents the most vulnerable segment of consumers: guardians responsible for family members with specific health risks. A

representative subject for this study is a single mother of three, including two children who suffer from severe allergies. Like many consumers, she faces a significant challenge in monitoring the constant stream of food recalls and frequently misses critical safety alerts from government sources like the FDA. This lack of real-time information leaves her family exposed to preventable health risks.

To test the efficacy of the solution, the case study will simulate a typical usage scenario where the user relies on Spoiler Alert to manage household food safety. In this scenario, the user will manually input her purchased groceries, such as chicken, into her personalized digital inventory immediately after shopping. The study will evaluate the system's ability to monitor these logged inventory items against emerging recall data in real-time.

The critical test of the application's value occurs when a hazard is detected. For example, if the chicken is flagged for salmonella contamination shortly after purchase, the app must successfully trigger an urgent notification. By receiving this alert before the food is prepared or consumed, the user can discard the product safely. This case study aims to demonstrate that Spoiler Alert can effectively bridge the gap between official recall notices and consumer awareness, providing a proactive defense through inventory monitoring.

4. Spoiler Alert Product Prototype Description

The Spoiler Alert prototype is designed to demonstrate the core functionality of the product: bridge the gap between official food recall notices and consumer awareness. While the real-world product (RWP) envisions a fully automated system with receipt and barcode scanning capabilities, the prototype will focus on proving the effectiveness of the notification engine and inventory management system. By streamlining the input process to manual entry, the prototype will sufficiently demonstrate how Spoiler Alert protects vulnerable consumers from unsafe food products without the overhead of implementing complex computer vision features in the initial phase.

4.1 Prototype Architecture

4.2 Prototype Features and Capabilities

Table 1: Feature Table

Category	Feature	RWP	Prototype
Account Creation	Account Registration with Email	☑	☑
	Account Signup with Facebook or Google	☑	✗
	Employee Account Creation	☑	☑
Home Page	Recall Timeline	☑	☑
	Newsfeed	☑	✗
Filtering	Filter Recalls Received	☑	☑
Item Tracking	Inventory Viewing	☑	☑
	Inventory Item Removal	☑	☑
	Barcode Scanning	☑	✗
	Receipt Scanning	☑	✗
Searching	Manual Product Entry	☑	☑
	Keyword Recall Search	☑	☑
Notifications	Push Notifications	☑	✗
	Email Alerts	☑	☑
	Missing a UPC Notification	☑	⚠
History	Recall History View	☑	☑

Recall Feed	Recent Recalls List	☒	☒
	Recall Details	☒	☒
Recall Reviewer Dashboard	Recall UPC Linking	☒	☒
	Manual Recall Editing	☒	☒

4.3 Prototype Development Challenges

5. Glossary

- Product Recall
- Recall Reviewer
- FDA
- CDC
- UPC
- CPSC
- API
- Allergen
- Immunocompromised
- Contamination

6. References

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